

Self-Test Mini Exercises

Instructions. Answer in full sentences unless you are asked for code or a short list. Where helpful, explain not only what a prior does, but also why it is appropriate for that parameter class and outcome scale.

Part I. Core understanding

Exercise 1: What changes when a model becomes Bayesian?

- In your own words, explain what people usually mean when they say they “turn a frequentist model into a Bayesian model.”
- Identify the roles of the likelihood, the prior, and the posterior.

Exercise 2: Match the prior to the parameter class

- For each parameter class below, state what kind of quantity it represents and name one prior family that is commonly used for it in brms:
 - a) class = "Intercept"
 - b) class = "b"
 - c) class = "sd"
 - d) class = "sigma"
- Then explain why the same prior would not automatically make equal sense across all four classes.

Exercise 3: Priors

- What is a Cauchy prior?
- How is it related to the Student-t family?
- Why do people describe it as “heavy-tailed”?
- Give one situation in which a Cauchy prior may be attractive and one situation in which it may be a poor default choice.

Exercise 4: Half-Cauchy and standard deviations

- Explain why a prior such as $\text{cauchy}(0, 3)$ on a standard deviation parameter behaves like a half-Cauchy in practice.
- Why are priors on class = "sd" especially important in multilevel models?
- What can happen if the prior on a random-effect SD is too narrow?

Exercise 5: SESOI

- In your own words, define SESOI.
- Explain the difference between statistical significance and practical importance.
- Explain the difference between a point null of exactly 0 and an interval null around 0.
- A posterior for a treatment effect is centered at 0.03, and your SESOI is $[-0.10, 0.10]$. What is the first practical question you should ask?
- A researcher says, “Any effect above zero matters.” What is wrong with that claim?
- Why is SESOI always tied to a scale, a context, and a decision problem?

Part II. Interpretation and diagnosis

Exercise 6: Scale matters

- Suppose you are modeling reaction times in milliseconds. Compare the practical meaning of the following two priors on `class = "sd"`:
 - a) `cauchy(0, 3)`
 - b) `cauchy(0, 100)`
- Discuss what each prior says before seeing the data, and why one might be too narrow or too diffuse depending on the scale of the outcome.

Exercise 7: Weakly informative is not the same as flat

- What is a weakly informative prior?
- Why is “as flat as possible” often not the best strategy?
- Give one example of how a weakly informative prior can help with estimation, interpretation, or computation.

Exercise 8: Matching priors to parameters

- Which parameter classes in a multilevel model usually need different priors: intercepts, slopes, residual SDs, random-effect SDs, or correlations? Explain briefly why they cannot all share the same prior.
- Why is a prior for a standard deviation usually chosen from a positive-only family or interpreted as truncated/half-distributed?
- Why should an intercept prior usually be chosen on the outcome scale rather than copied from a prior used for standardized slopes?

Exercise 9: Reading results against a SESOI

- Suppose an effect estimate is 0.18 with a 95% credible interval of [0.02, 0.31], and the SESOI is [-0.10, 0.10]. Write 3-4 sentences interpreting this result from a practical-importance perspective.
- Now suppose an effect estimate is 0.05 with a 95% credible interval of [-0.02, 0.12], with the same SESOI. What is uncertain here? Be precise.
- Now suppose an effect estimate is 0.01 with a 95% credible interval of [-0.03, 0.04], again with the same SESOI. What would be the most defensible conclusion?
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Part III. Applied work in R

Exercise 10: Writing priors in brms

- Write a brms prior specification for a Gaussian mixed model with these choices:
 - a normal prior for the intercept centered on 750 with scale 200
 - a normal prior for fixed effects centered on 0 with scale 50
 - a Student-t prior for random-effect standard deviations with $df = 3$, location = 0, scale = 50
 - a Student-t prior for sigma with $df = 3$, location = 0, scale = 50
 - Write the R code using `prior(...)` or `set_prior(...)`.

Exercise 11: Inspecting defaults before overriding them

- What function would you use in brms to inspect priors before fitting a model?
- Why is this a good habit?
- Write a short code example using a formula of your choice.

Exercise 12: Sensitivity analysis

- Imagine you fit the same model twice. In Model A you use a narrow prior on class = "sd"; in Model B you use a much wider prior. The posterior intervals for the main effects are noticeably wider in Model B.
- Give two possible interpretations of this result.
- Then explain what you would report in a write-up to show that you took prior sensitivity seriously.

BONUS (criticize, criticize, criticize)

1. A student writes: "We used a very wide prior because wide priors are always more objective." Critique this statement.
2. Another student writes: "Our estimate was statistically different from zero, so it must be practically meaningful." Critique this statement.
3. A third student writes: "We did not have time to read the literature, so we chose a SESOI randomly." Rewrite this into a more defensible sentence.

Optional challenges

Challenge 1: Prior justification

Write a short methods paragraph justifying a prior choice for a Bayesian mixed model of reaction times. Your paragraph should mention the outcome scale, the parameter class, and why the chosen prior is reasonable rather than arbitrary.

Challenge 2: Prior predictive thinking

Without fitting a model, explain what kinds of implausible predictions would make you distrust a prior for a reaction-time model.

Be specific: what would count as “nonsense” on the outcome scale, and how could prior predictive checking help you catch it?

Challenge 3: SESOI paragraphs

Write a short methods paragraph that justifies a provisional SESOI for a classroom Bayesian analysis where time was limited.

Write a second paragraph justifying one prior for fixed effects and one prior for random-effect SDs.

Your explanation should mention scale, plausibility, and uncertainty without pretending the choices are uniquely correct.